

ServiceNow-Based Employee Laptop Request Application

Project Documentation

Platform: ServiceNow (Scoped Application)

Module Type: Service Catalog / Record Producer / Flow Designer

1. Introduction

Project Title

ServiceNow-Based Employee Laptop Request Application

Team Members

[Add team member names and roles here]

Platform Overview

This project is built entirely on the ServiceNow platform using a custom scoped application (**RR application**, scope **x_2065290_rr_app_0**). Unlike a traditional MERN-stack build, the 'frontend', 'backend', and 'database' layers here are all native ServiceNow constructs: Service Catalog / Record Producer for the UI, Flow Designer and Business Rules for server-side logic, and a custom ServiceNow table for data storage.

2. Project Overview

Purpose

The Employee Laptop Request Application allows employees to raise a request for a new laptop directly from a ServiceNow catalog form. The request automatically routes to the IT Manager group for approval, and the requester and manager are notified by email at each stage. The goal is to digitize and standardize what would otherwise be an ad-hoc, email-based laptop request process.

Key Features

- A Record Producer ("Employee laptop request") that presents a simple catalog form to employees.
- Custom table **Laptop Request Form** (`u_laptop_request_form`) that stores every request record.
- An approval workflow built in Flow Designer that asks the IT Manager group to approve or reject each request.
- Conditional branching in the flow based on the *Type of Laptop* (e.g. Basic vs Advance) selected by the employee.
- Automatic email notifications on approval and rejection.
- A Business Rule that prevents an employee from submitting more than one laptop request in the same calendar year.
- Role-based security using two custom roles (*it employees*, *it manager*) and two groups (*it employees*, *it managers*) enforced through Access Control Rules (ACLs).
- A Relationship record that surfaces the Approval history directly on the Laptop Request Form.

3. Architecture

The application follows ServiceNow's standard three-layer model. The table below maps each MERN concept to its ServiceNow equivalent used in this project.

Layer	ServiceNow Equivalent
Frontend (React)	Service Catalog Item / Record Producer + native Form/List UI
Backend (Node.js / Express)	Flow Designer flow, Business Rules, Relationship scripts
Database (MongoDB)	Custom table: <code>u_laptop_request_form</code> (and related <code>sys_approval_approver</code>)

3.1 Scoped Application

All components (table, record producer, flow, business rule, roles, groups and ACLs) live inside a single custom scoped application called **RR application** (scope prefix `x_2065290_rr_app_0`), created from ServiceNow Studio. Building in an application scope keeps the tables and logic isolated from the Global scope and makes the whole application easy to move between instances as one Update Set.

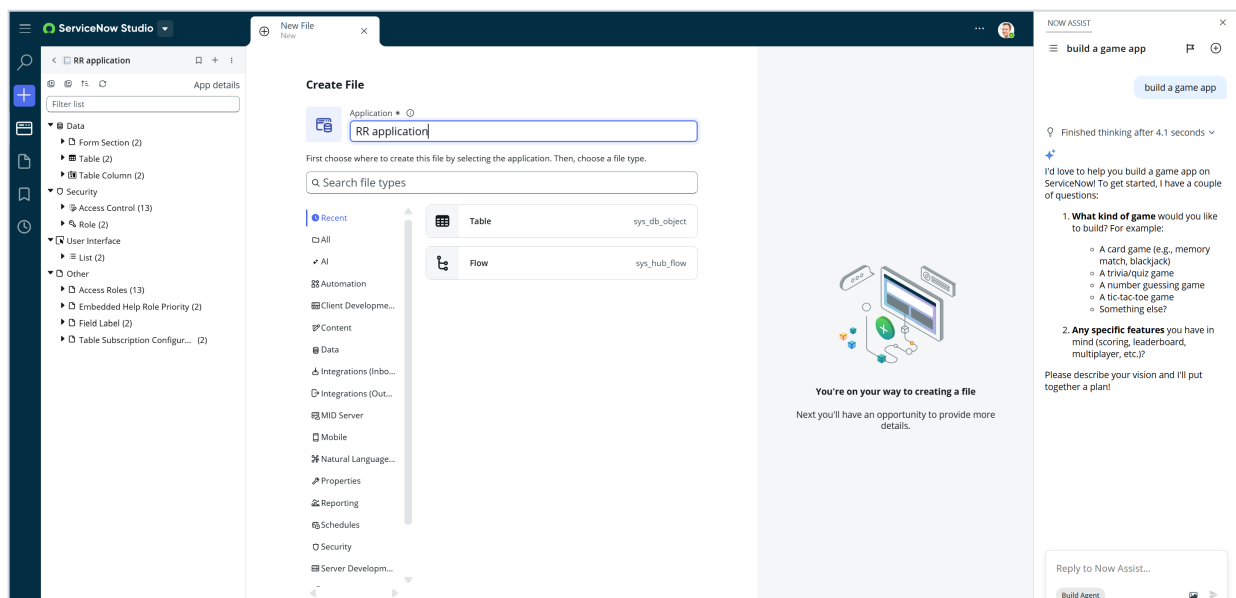


Figure 1: Creating a new file inside the RR application in ServiceNow Studio. Now Assist (right panel) can also generate application artifacts from a natural-language description.

3.2 Data Model — Laptop Request Form Table

The core data store is the custom table **Laptop Request Form**. Each record captures who is requesting a laptop, for whom, what type, and the current approval status.

Field (Question)	Column Name	Type
What is employee name	u_requester	Reference (sys_user)
Employee mail id	u_email_2	Email
When did you want	u_date	Date/Time
Short description	u_short_description	Multi-Line Text
Type of laptop	—	Multiple Choice
status	u_status	Multiple Choice
Reason	u_reason	Multiple Choice

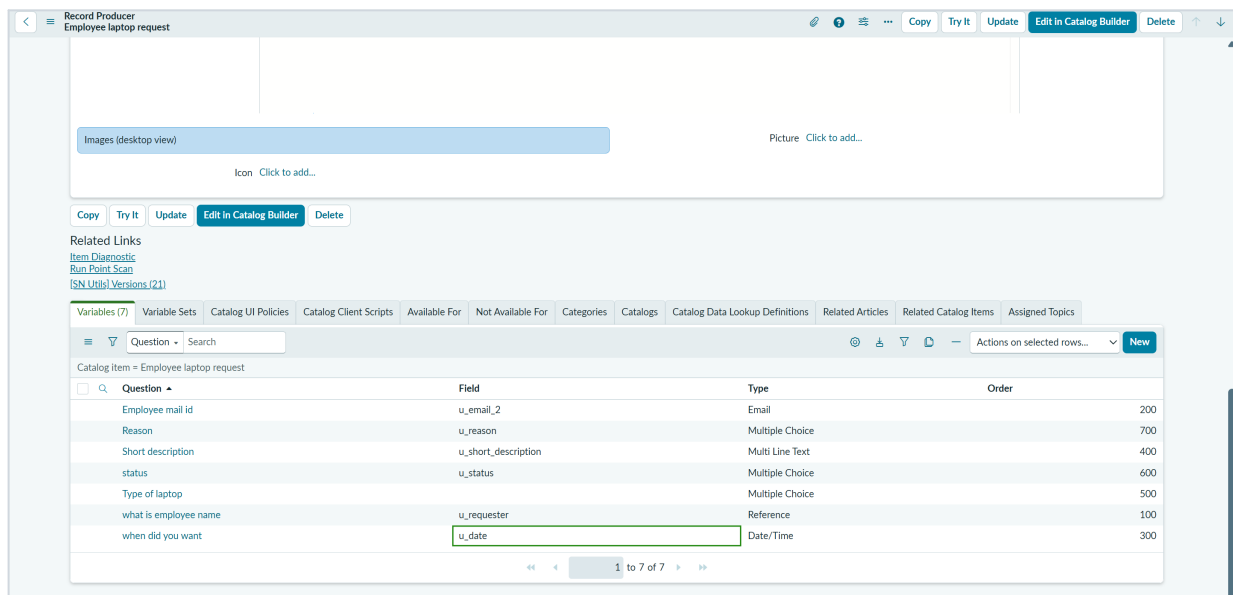


Figure 2: The 7 variables/fields defined on the Employee laptop request Record Producer, which map directly onto columns of the u_laptop_request_form table.

3.3 Record Producer

A **Record Producer** is the ServiceNow equivalent of a public-facing intake form. The "Employee laptop request" record producer collects the employee's name, email, manager, preferred date, laptop type, and reason, and on submission creates a new record in the u_laptop_request_form table.

3.4 Flow Designer — Approval Workflow

The **Laptop Request Flow** is triggered whenever a new Laptop Request Form record is created. It first checks the requested *Type of Laptop* using an If condition, then routes the record for approval and finally updates the record and sends an email depending on whether the request is approved or rejected.

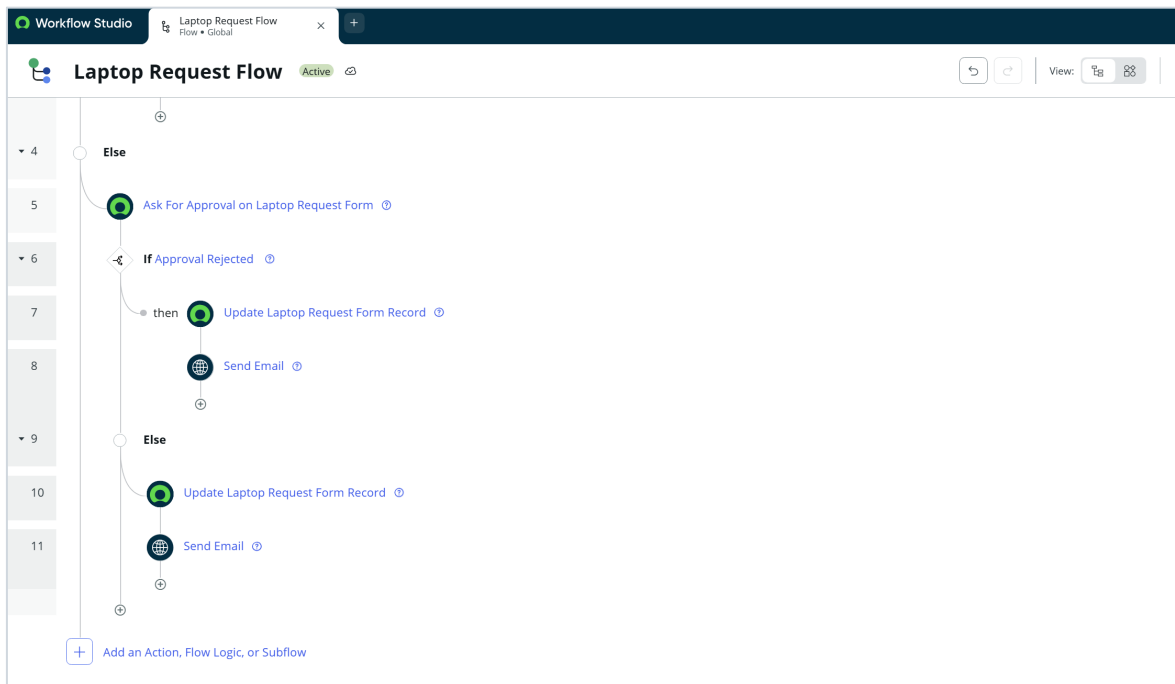


Figure 3: Flow trigger ("Laptop Request Form Created") and the first If condition, labelled "Laptop is Basic", which checks whether Type of Laptop equals "basic".

Number: LAP0001014
 Requester: Abel Tutor
 Employee Email: abel.tutor@example.com
 Date: 2026-07-20
 Short Description:
 Manager: Sabrina Deppert
 Type of Laptop: Advance
 Status: requested
 Reason: -- None --

Update Delete

State	Approver	Comments	Approval for	Created
Requested	(empty)		(empty)	2026-07-29 21:41:01

1 to 1 of 1

Figure 4: Else branch of the flow: an approval is requested on the Laptop Request Form record. If the approval is rejected, the record is updated and a rejection email is sent; otherwise (approved) the record is updated and an approval email is sent.

3.5 Business Rule — Duplicate Request Guard

A server-side **Business Rule** named "Laptop request BR" runs before insert and blocks an employee from submitting a second laptop request within the same calendar year. It queries the `u_laptop_request_form` table for existing requests from the same requester between the start and end of the current year, and aborts the action with an error message if one is found.

Figure 5: Advanced script of the "Laptop request BR" Business Rule. It uses `gs.beginningOfThisYear()` and `gs.endOfThisYear()` to look for a prior request from the same requester and calls `current.setAbortAction(true)` to block a duplicate submission.

3.6 Relationship — Approval History

A **Relationship** record called "Laptop request" links the Laptop Request Form table to the Approval (sysapproval_approver) table. Its `refineQuery` script filters approvals whose `source_table` and `document_id` match the current laptop request record, so every approval raised for a request shows up as a related list directly on the form.

Figure 6: The "Laptop request" Relationship record. Its query script joins the Approval table to the current Laptop Request Form record using `source_table` and `document_id`.

4. Setup Instructions

Prerequisites

- A ServiceNow instance (Personal Developer Instance or sandbox) with admin access.
- The **security_admin** role elevated for the admin account, required to create/modify Access Control Rules.
- Access to ServiceNow Studio and Workflow Studio (Flow Designer).

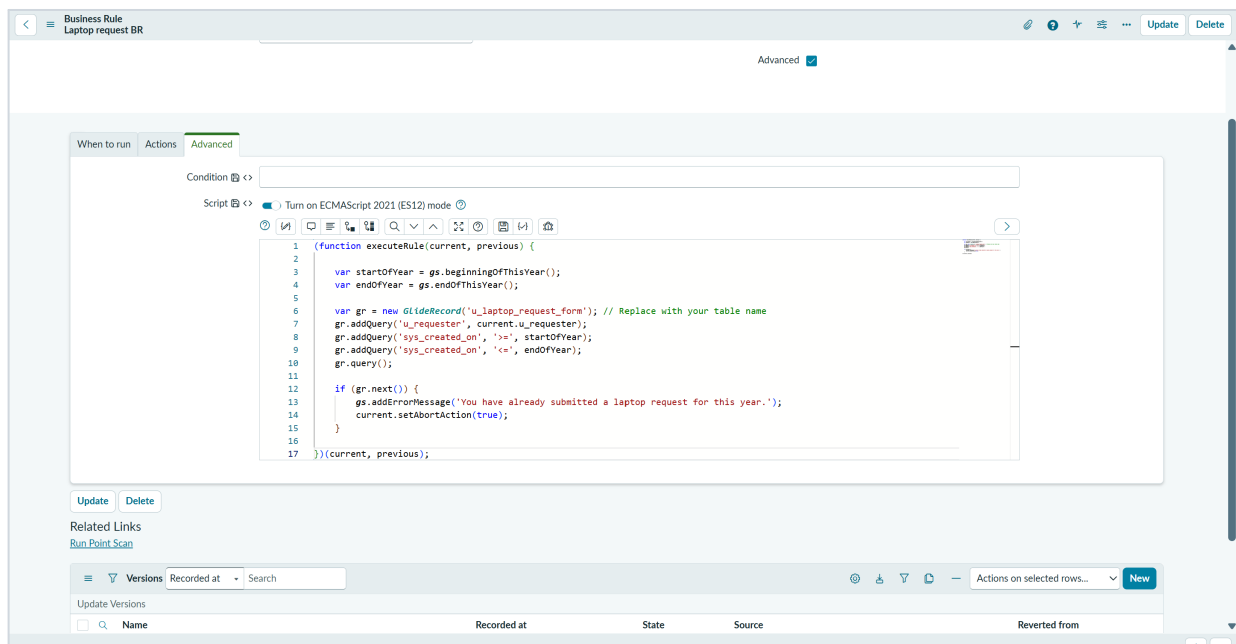


Figure 7: Elevating to the **security_admin** role from the user profile menu. This elevated privilege is required before Access Control Rules (ACLs) can be created or edited.

Installation — Importing the Update Set

All customizations for this project (table, record producer, flow, business rule, relationship, roles, groups and ACLs) are captured in a single Update Set named **"HR Onboarding Application"**. To install the application on another instance:

- On the source instance, open the Update Set and use **Export to XML** to download the remote update set XML file.
- On the target instance, navigate to *Retrieved Update Sets* and import the XML file.
- Open the imported Update Set and click **Preview Update Set**, resolve any conflicts, then click **Commit Update Set**.
- Verify that the RR application, the Laptop Request Form table, the Record Producer, the Flow, the Business Rule, the Roles/Groups and the ACLs have all been created.

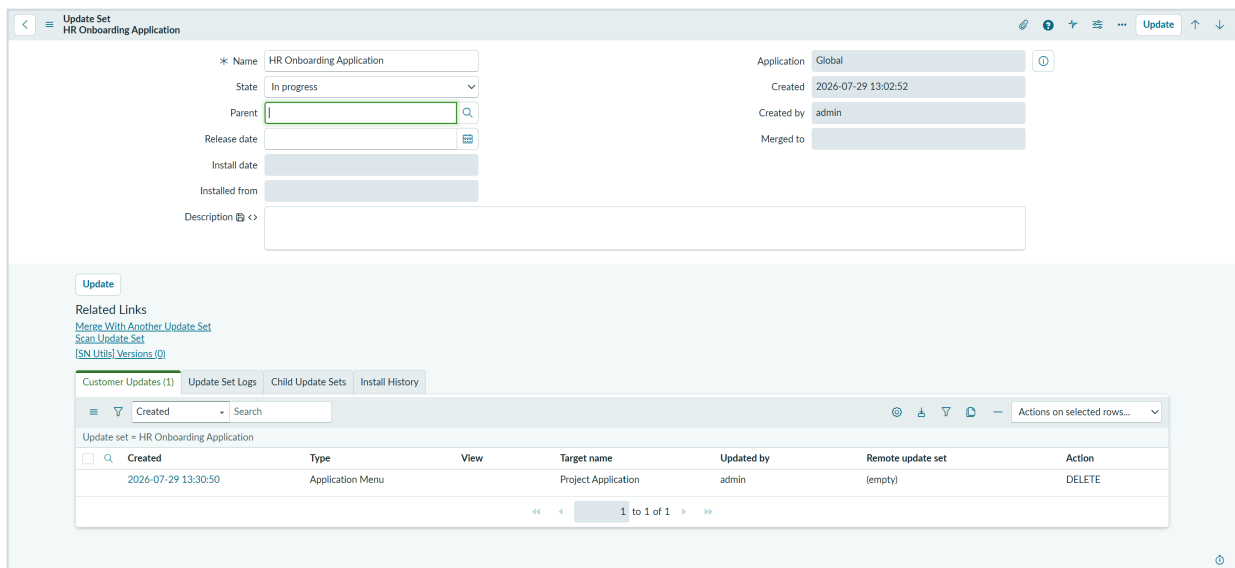


Figure 8: The "HR Onboarding Application" Update Set in "In progress" state, showing the list of customer updates captured for this project.

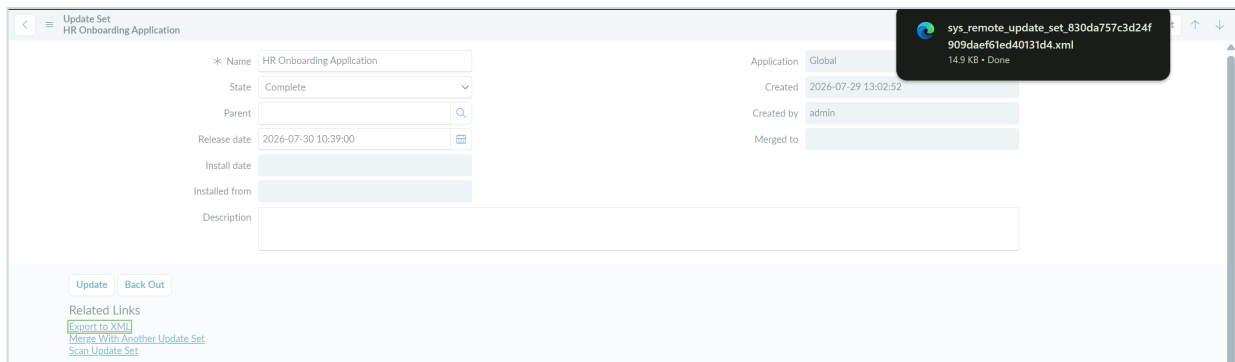


Figure 9: Exporting the completed Update Set to an XML file using the "Export to XML" related link, ready to be imported into another ServiceNow instance.

5. Application Structure

Instead of a client/server folder structure, a ServiceNow scoped application organizes its components by type inside Studio. The main components created for this project are:

Component	Name / Table	Purpose
Table	Laptop Request Form (u_laptop_request_form)	Stores every laptop request record
Record Producer	Employee laptop request	Catalog form used by employees to raise a request

Component	Name / Table	Purpose
Flow	Laptop Request Flow	Approval routing and notification logic
Business Rule	Laptop request BR	Blocks duplicate requests in the same year
Relationship	Laptop request	Shows related Approval records on the form
Roles	it employees, it manager	Custom security roles
Groups	it employees, it managers	Assignment/approval groups
Access Controls	4 ACLs on Laptop Request Form	Read / Create / Write security

6. Running the Application

There is no separate frontend/backend server to start — the application runs natively inside the ServiceNow instance. An employee opens the "Employee laptop request" item from the Service Catalog (or its direct Record Producer URL), fills in the blank form, and clicks Submit to create a new Laptop Request Form record.

Figure 10: The blank Laptop Request Form as presented to the employee before submission, with fields for Number, Requester, Employee Email, Date, Type of Laptop, Status and Reason.

Once submitted, the record is created with a Status of "requested" and the Flow automatically raises an Approval record. Both the request record and its approval history are visible together on the form, thanks to the Relationship configured in Section 3.6.

Figure 11: A submitted Laptop Request Form record (LAP0001014) showing the requester, manager, laptop type and the related Approval record in "Requested" state at the bottom of the form.

7. Data / API Documentation

ServiceNow does not require hand-written REST endpoints for basic CRUD — every table automatically exposes a REST API (the Table API) at **/api/now/table/u_laptop_request_form**. The table below documents the operations available and the equivalent UI action that triggers them in this application.

Operation	Table API Method	Triggered From
Create a request	POST /api/now/table/u_laptop_request_form	Record Producer submission
Read a request	GET /api/now/table/u_laptop_request_form/{sys_id}	Opening the Laptop Request Form record
Update a request	PUT /api/now/table/u_laptop_request_form/{sys_id}	Flow Designer "Update Laptop Request Form Record" action
List approvals	GET /api/now/table/sysapproval_approver	Approvals related list (via Relationship)

Access to every one of these operations is still governed by the Access Control Rules described in Section 8 — the Table API enforces the same role checks as the UI.

8. Authentication & Authorization

Instead of JWTs or sessions managed in application code, this project relies entirely on ServiceNow's built-in platform security: native user sessions, custom Roles, Groups, and Access Control Rules (ACLs) on the Laptop Request Form table.

8.1 Roles

Two custom roles were created to distinguish employees who can raise a request from managers who can approve one.

The screenshot shows the 'Group' configuration page for 'it managers'. At the top, there's a notification bar: 'Job to add or remove role(s) from user(s) of group has been queued'. Below this, the 'Name' field is 'it managers', 'Group email' is empty, 'Manager' is empty, and 'Parent' is empty. The 'Description' field contains 'who approves the request'. There are 'Update' and 'Delete' buttons. Below the form, there's a section for 'Roles (1)' and 'Group Members (8)'. The 'Group Members' tab is active, showing a list of users: Byron Fortuna, Ashley Leonesio, Bryan Rovell, Cary Mccamey, Bow Ruggeri, Bette Barcelona, Benchmark Scheduler, and Candice Bruckman. The list is paginated, showing '1 to 8 of 8'.

Figure 12: The custom "it employees" role, granted to staff who are allowed to submit laptop requests.

The screenshot shows the 'Role' configuration page for 'it employees'. The 'Name' field is 'it employees', 'Application' is 'Global', and 'Elevated privilege' is unchecked. The 'Description' field is empty. There are 'Update' and 'Delete' buttons. Below the form, there's a section for 'Contains Roles', 'Applications with Role', 'Modules with Role', and 'Custom Tables'. The 'Contains Roles' tab is active, showing a search bar with 'for text' and a search button. Below the search bar, there's a message: 'No records to display'.

Figure 13: The custom "it manager" role, granted to IT managers who approve or reject laptop requests.

8.2 Groups

Two groups mirror the two roles and are used for assignment and approval routing. The "it managers" group is the group the Flow asks for approval; the "it employees" group represents the staff who are entitled to submit requests.

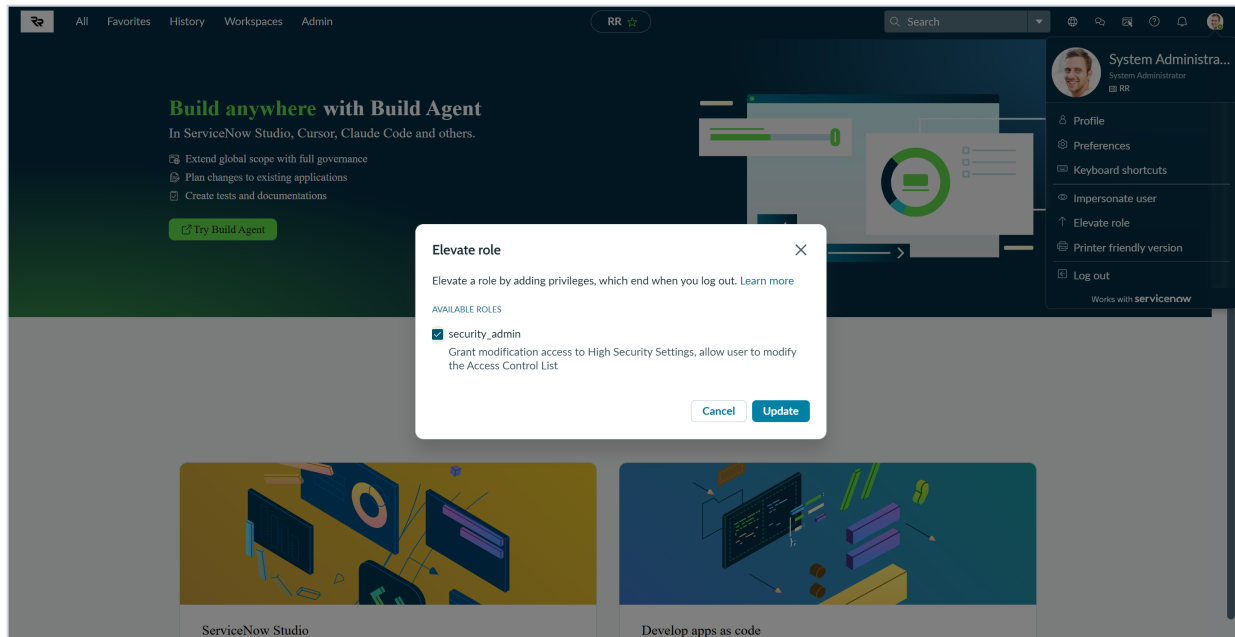


Figure 14: The "it managers" group (8 members), described as "who approves the request" — this is the approval group used by the Laptop Request Flow.

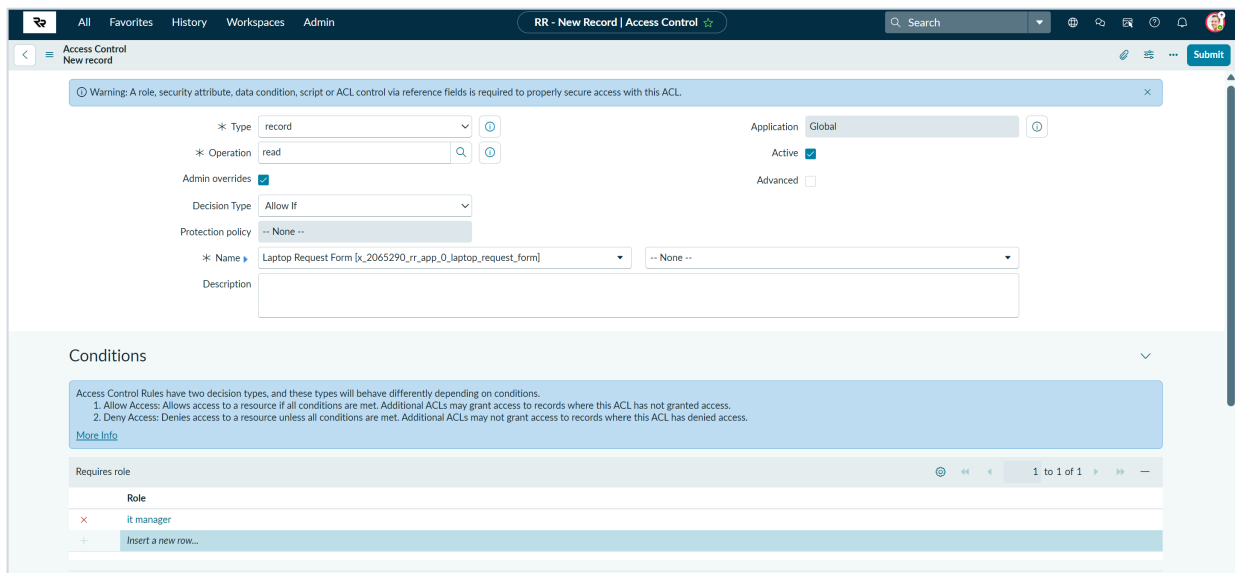


Figure 15: The "it employees" group (10 members), described as "Employee members in it companies".

8.3 Access Control Rules (ACLs)

Four Access Control Rules on the `u_laptop_request_form` table enforce role-based security: an employee (role *it employees*) can create and read/write their own requests, while a manager (role *it manager*) can read requests in order to approve them.

The screenshot shows the 'Access Control' configuration page for the table 'x_2065290_rr_app_0_laptop_request_form'. The configuration is for a 'record' type with the 'Operation' set to 'create'. The 'Application' is 'Global', and the 'Active' checkbox is checked. The 'Admin overrides' checkbox is also checked. The 'Decision Type' is set to 'Allow If'. The 'Protection policy' is set to '-- None --'. The 'Name' is 'Laptop Request Form [x_2065290_rr_app_0_laptop_request_form]' and the 'Description' is 'Allow create for records in x_2065290_rr_app_0_laptop_request_form, for users with role it employees.'.

Conditions

Access Control Rules have two decision types, and these types will behave differently depending on conditions.

1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

Role
it employees

1 to 1 of 1

Figure 16: A new "read" Access Control record being created on the Laptop Request Form table, restricted to users with the "it manager" role.

The screenshot shows the 'Access Control' configuration page for the table 'x_2065290_rr_app_0_laptop_request_form'. The configuration is for a 'record' type with the 'Operation' set to 'read'. The 'Application' is 'Global', and the 'Active' checkbox is checked. The 'Admin overrides' checkbox is also checked. The 'Decision Type' is set to 'Allow If'. The 'Protection policy' is set to '-- None --'. The 'Name' is 'Laptop Request Form [x_2065290_rr_app_0_laptop_request_form]' and the 'Description' is empty.

Conditions

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1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role

Role
it employees

1 to 1 of 1

Figure 17: The "create" Access Control on Laptop Request Form, restricted to the "it employees" role — "Allow create for records in x_2065290_rr_app_0_laptop_request_form, for users with role it employees."

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1. Allow Access: Allows access to a resource if all conditions are met. Additional ACLs may grant access to records where this ACL has not granted access.
2. Deny Access: Denies access to a resource unless all conditions are met. Additional ACLs may not grant access to records where this ACL has denied access.

[More Info](#)

Requires role
Role it employees

Figure 18: The "read" Access Control on Laptop Request Form, also restricted to the "it employees" role, so requesters can view their own submitted records.

Workflow Studio Laptop Request Flow Flow • Global

Laptop Request Flow Active

TRIGGER

Laptop Request Form Created

ACTIONS Select multiple

1 If

Condition Label: Laptop is Basic

* Condition 1: Trigger - R... > ... > Type of La... is basic

Add another condition set(OR)

Figure 19: The "write" Access Control on Laptop Request Form, restricted to the "it employees" role — "Allow write for records in x_2065290_rr_app_0_laptop_request_form, for users with role it employees."

Operation	Table	Required Role
create	Laptop Request Form	it employees
read	Laptop Request Form	it employees
write	Laptop Request Form	it employees
read	Laptop Request Form	it manager

9. User Interface

The end-user interface is entirely native ServiceNow UI — no custom frontend code was written. Two surfaces make up the user experience:

- **Employee-facing:** the "Employee laptop request" Record Producer form (Section 6), used to submit a new request.
- **Manager/admin-facing:** the standard ServiceNow record view of the Laptop Request Form table, which shows the request details together with its live Approval status via the related list (Section 3.6).

Because the form is built with out-of-the-box ServiceNow UI elements (reference fields, choice lists, date pickers), it automatically inherits the platform's responsive layout, accessibility and theming.

10. Testing

Testing for this application was carried out manually within the ServiceNow instance, using the following approach:

- **Form submission testing:** submitting the Employee laptop request form with valid and missing field values to confirm mandatory fields are enforced.
- **Duplicate-request testing:** submitting a second laptop request for the same employee within the same year to confirm the "Laptop request BR" Business Rule blocks it with the expected error message.
- **Flow testing:** creating requests with different "Type of Laptop" values to confirm the "Laptop is Basic" condition branches correctly, and approving/rejecting the resulting approval record to confirm the record status updates and the correct email is sent.
- **Security testing:** logging in as a user with only the "it employees" role to confirm they can create and view their own requests but cannot access manager-only functions, and as a user with the "it manager" role to confirm they can read requests routed to them for approval.
- **Update Set validation:** exporting the Update Set to XML and re-importing it into a clean instance to confirm every component (table, record producer, flow, business rule, relationship, roles, groups, ACLs) migrates correctly.

11. Screenshots / Demo Summary

Screenshots for every major component are embedded throughout this document, in the section that describes the corresponding feature (Architecture, Setup, Running the Application, and Authentication & Authorization). The quick index below lists every figure for reference.

Fig.	Shows
1	RR application — creating a file in ServiceNow Studio (Now Assist panel)
2	Record Producer variables for the Employee laptop request form

Fig.	Shows
3	Flow trigger and "Laptop is Basic" If condition
4	Flow approval branch — approve/reject actions
5	"Laptop request BR" Business Rule script
6	Relationship record joining Approval to Laptop Request Form
7	Elevating to the security_admin role
8	"HR Onboarding Application" Update Set — Customer Updates
9	Exporting the Update Set to XML
10	Blank Laptop Request Form (employee view)
11	Submitted Laptop Request Form record with Approval related list
12	"it employees" role
13	"it manager" role
14	"it managers" group (approval group)
15	"it employees" group
16	Read ACL scoped to "it manager" role
17	Create ACL scoped to "it employees" role
18	Read ACL scoped to "it employees" role
19	Write ACL scoped to "it employees" role

12. Known Issues

- The *Type of Laptop* variable on the Record Producer does not have a backing field set on the Variables list, which should be mapped to a table column so its value is queryable in reports.
- The Business Rule's duplicate-request check is scoped to the calendar year only; it does not account for a request that is rejected — a rejected employee currently cannot raise a new request until the following year.
- Email notifications are sent via a generic "Send Email" flow action; sender address and templates have not yet been finalized/branded.
- The Manager field on the form is a free-text reference lookup and is not currently auto-populated from the requester's manager field on the user record.

13. Future Enhancements

- Auto-populate the Manager field from the requester's sys_user.manager record.
- Add a Service Portal widget so employees can track the live status of their request without opening the platform UI.
- Extend the Flow to integrate with Asset Management so an approved request automatically creates a hardware asset assignment task.
- Add a Performance Analytics / reporting dashboard summarizing requests by laptop type, status and turnaround time.
- Allow a rejected request to be resubmitted with manager comments instead of blocking further requests until the next calendar year.
- Add mobile-friendly UI Actions so managers can approve or reject requests directly from the ServiceNow mobile app.